05 Primary Model - Dynamic Discrete-Time Competing-Risks Hazard Model

Abdul Ahad. Student ID: G00486649

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0.1 Introduction

The primary model is a **multinomial pooled logistic regression** (D'Agostino et al., 1990; Heyard et al., 2019). If this interpretable regression matches gradient-boosted trees (H6), black-box complexity is unnecessary.

For patient i at hour t, with competing events $r \in \{1{=}\text{sepsis}, 2{=}\text{discharge}, 3{=}\text{death}\}$:

$$\lambda_r(t \mid X_{it}) = \frac{\exp(\alpha_r(t) + \beta_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \beta_s^\top \mathbf{X}_{it})}$$

$\alpha_r(t)$ is the baseline log-hazard (natural cubic spline in hours since admission) and β_r are cause-specific covariate coefficients. Standard errors use a cluster-robust sandwich estimator (clustered on `stay_id`).

Models are fitted per label variant: Variant B is primary; A and C support the cross-label coefficient comparison (H2). The alternative-label variants of Protocol Amendment 2 (D, E) are fitted with exactly the same specification so that the label is the only thing that differs; their coefficients are kept in a separate file and take no part in H2.

0.2 Feature Set

Predictors include vitals, lab values, vasopressor exposure, and 6-hour rolling slopes. Missingness indicators let the model distinguish normal values from unmeasured ones.

```
## Base features: hr, resp_rate, spo2, map, temp_c, gcs, lactate, creatinine, bilirubin_total, p
```

```
## Missingness indicators: lactate_measured, creatinine_measured, bilirubin_total_measured, plat
```

0.3 Fit Primary Model Per Variant

Per variant: load training person-hours, impute remaining NAs to zero, fit the multinomial model via `nnet::multinom`, compute cluster-robust SEs, predict on the test set, and save outputs. A binary logistic fallback handles non-convergence.

```
##   Training rows: 2,304,923, events (sepsis): 4647
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 4, 13, 24, 38, 58
##   Fit diagnostics [A]: convergence=0  deviance=459935.5  max|coef|=12.871
##   Variant A model saved (arm = full).
##   Training rows: 2,301,842, events (sepsis): 4847
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 4, 13, 24, 38, 58
##   Fit diagnostics [B]: convergence=0  deviance=462017.5  max|coef|=12.262
##   Variant B model saved (arm = full).
##   Training rows: 2,120,376, events (sepsis): 9403
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 4, 12, 22, 36, 57
##   Fit diagnostics [C]: convergence=0  deviance=474892.4  max|coef|=14.131
##   Variant C model saved (arm = full).
##   Training rows: 1,601,145, events (sepsis): 22435
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 3, 9, 18, 31, 53
##   Fit diagnostics [D]: convergence=0  deviance=463984.8  max|coef|=18.857
##   Variant D model saved (arm = full).
##   Training rows: 1,927,446, events (sepsis): 14178
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 3, 11, 21, 34, 55
##   Fit diagnostics [E]: convergence=0  deviance=479315.2  max|coef|=15.909
##   Variant E model saved (arm = full).
##   Training rows: 2,275,961, events (sepsis): 4992
##   Feature check: all 22 declared features present.
##   Spline knots at hours: 4, 13, 23, 38, 58
##   Fit diagnostics [F]: convergence=0  deviance=460612.3  max|coef|=12.658
##   Variant F model saved (arm = full).

##   Training rows: 2,304,923, events (sepsis): 4647
##   Feature check: all 22 declared features present.
##   Ablated arm: 8 action-derived features dropped, 14 remain
##   Spline knots at hours: 4, 13, 24, 38, 58
##   Fit diagnostics [A]: convergence=0  deviance=463538.2  max|coef|=12.031
##   Variant A model saved (arm = ablated).
##   Training rows: 2,301,842, events (sepsis): 4847
##   Feature check: all 22 declared features present.
##   Ablated arm: 8 action-derived features dropped, 14 remain
##   Spline knots at hours: 4, 13, 24, 38, 58
##   Fit diagnostics [B]: convergence=0  deviance=465678.8  max|coef|=11.891
##   Variant B model saved (arm = ablated).
##   Training rows: 2,120,376, events (sepsis): 9403
```

Table 1: Top 15 sepsis-outcome coefficients (Variant B, by |coef|), with cluster-robust standard errors

term	coef	se_cluster	variant
platelets_measured	-0.5613720	0.0445148	B
wbc_measured	0.5373895	0.0432729	B
bilirubin_total_measured	0.4386618	0.0346256	B
norepi_epi_anyTRUE	0.3601162	0.0690386	B
lactate_measured	0.3504400	0.0413008	B
creatinine_measured	0.2303394	0.0876647	B
pf_ratio_measured	0.1054718	0.0439808	B
creatinine	0.0514024	0.0092022	B
lactate	-0.0340341	0.0098207	B
vaso_anyTRUE	0.0270614	0.0601292	B
resp_rate	0.0218536	0.0033374	B
gcs	-0.0200954	0.0034209	B
temp_c	0.0133121	0.0021250	B
hr	0.0125341	0.0010504	B
resp_rate_slope6h	0.0083772	0.0047949	B

```
##   Feature check: all 22 declared features present.
##   Ablated arm: 8 action-derived features dropped, 14 remain
##   Spline knots at hours: 4, 12, 22, 36, 57
##   Fit diagnostics [C]: convergence=0  deviance=478825.9  max|coef|=13.202
##   Variant C model saved (arm = ablated).
```

0.4 Random-Split Re-Fit (H4: Split Optimism)

The same primary model is re-fitted on a random patient-level split (Guo et al., 2022). Only Variant B is needed: H4 compares temporal vs. random AUROC under a fixed label and model class.

```
##   Random-split training rows: 2,140,496, events (sepsis): 4423
##   Feature check: all 22 declared features present.
##   Fit diagnostics [B (random split)]: convergence=0  deviance=425013.3  max|coef|=12.683
##   Random-split model for Variant B saved.
```

0.5 Cross-Label Coefficient Stability

The combined coefficient table across variants feeds the H2 (coefficient instability) test: if a covariate changes sign across label variants, the label choice qualitatively alters the model's clinical logic.

```
## Cross-label coefficient tables saved for H2 testing.
```

```
## Alternative-label coefficient tables saved (Protocol Amendment 2).
```

0.5.1 Ablated-Arm Stability (Protocol Amendment 5)

The same instability rule applied to the ablated fit. The question is whether the cross-label instability observed in the full specification is a property of the model as a whole or specifically of the features that record clinician behaviour: if the latter, removing those features should leave a markedly more stable set of coefficients behind.

Table 2: Unstable covariates remaining after ablation

term	sign_flip	mag_flip
lactate	FALSE	TRUE

```
## Ablated arm: of 14 shared covariates, 0 reverse sign, 1 change magnitude >=2x, 1 do either
```